

Fig. 11: Actual-size PCB of the touch-control module

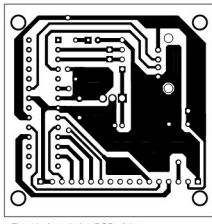


Fig. 13: Actual-size PCB of the keyboard module

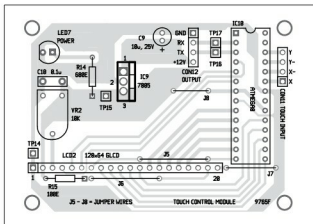


Fig. 12: Component layout of the touch-control module

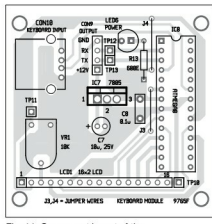


Fig. 14: Component layout of the keyboard module

EFY Note

The source codes of this project are included in this month's EFY DVD and are also available for free download at source.efymag.com

`readm()` is used for module name verification.

All functions related to LCD operation are included in `lcd.h` header file.

Construction and testing

Actual-size, single-side PCBs for the four modules are shown in Figs 7, 9, 11 and 13, while their respective component layouts are shown in Figs 8, 10, 12 and 14.

Before using any SMPS power supply, you need to check ATX specifications, key mechanical dimensions, mounting points, I/O panels,

power and connector interfaces. You also need to check SMPS connector pin details properly before connecting to the circuit.

There are 20-pin, 6-pin and 4-pin output connectors available on SMPS for various sections in the CPU. Wires are normally colour-coded; yellow is 12V, red is 5V, orange is 3.3V, black is ground and green is for PS_ON (power on). You need to identify these wires, connect to suitable connectors for use in this circuit. Green wire should be connected to ground through power switch (S1) on the front panel of the CPU.

When AC supply is available on SMPS, it activates the +5V_{SB} power supply on the main module, and the system goes to standby mode. As soon as the power switch (PS_ON) is pressed, the main module becomes active and power-hold relay RL2 is

energised, thus holding the supply while the program checks various statuses.

On completion of status checks, the system LED8 (RGB) will indicate the status of the load.

At this stage, unlock the touch-control panel by pressing Unlock, select the load or appliance you want to operate. Touch On or Off to switch on or switch off the load.

To change settings using the keyboard module, disconnect the touch-control module from CON4 and connect the keyboard module to it. Connect a PS/2 type keyboard to CON10 using a PS/2-to-USB converter. You will see various setting options on the LCD.

Testing of individual modules can be done by selecting the module and then the test mode. You can select the command from the command mode. Upon registering of the command, the system tests all devices' on/off status indicators (RGB LEDs) and their corresponding relays.

For troubleshooting, first check the voltages at various test points mentioned in Table II. Reset switch S2, which is connected on the front panel of the CPU, can be used during troubleshooting. On pressing switch S2, RL1 and RL2 are de-energised, and hence system goes to standby mode. ●

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